

Ajay Madana

College Park, MD (open to relocate) | ajaymadana2126@gmail.com | 240-879-8242
[linkedin.com/in/ajaymadana126](https://www.linkedin.com/in/ajaymadana126) | Portfolio | github.com/Ajay0612

Education

University of Maryland, College Park <i>College Park, MD</i>	MS in Data Science (4.0 GPA)	May 2027
- Member of BigthinkAI club at UMD - Coursework: Big Data Systems, Algorithms for Data Science, Data Representation & Modelling		
Amrita Vishwa Vidyapeetham, Coimbatore <i>Tamil Nadu, India</i>	B.Tech in Computer Science (8.2 GPA)	2021-2025
- Coursework: Data Structures & Algorithms, Time Series Forecasting, Business Analytics, Machine Learning, Deep Learning, Data Science		

Work Experience

YugaSys (HeiBuddy Startup App) - AI Product & Data Analyst Intern <i>Remote</i>	Jun 2026 - Present
- Translated planned AI architecture into measurable QA checks across 5 areas: 2-lane policy routing (sensitive doctor vs standard cab), DAG dependencies, async MCP tool execution, audit logs, and observability. - Built spreadsheet-based QA tracker categorizing defects across 4 impact dimensions: feature, severity, frequency, and user impact; used a 250 ms Graph RAG timeout target to quantify fallback behavior and reliability trade-offs.	
SAP - Intern, Team Lead <i>Bangalore, India</i>	Mar 2024 - Apr 2024
- Led a small intern project team to prototype an OCR-powered spend analysis workflow using AWS S3, Textract, and DynamoDB; benchmarked a 40% improvement in sample receipt-processing time versus manual review. - Designed a receipt sustainability scoring approach that classified purchased items, calculated the percentage of sustainable items per receipt, and converted the result into a user-facing sustainability score.	

Academic Projects

Medicare Spend Intelligence - Healthcare Analytics (<i>PostgreSQL, Advanced SQL, Data Quality, Healthcare Analytics</i>)	Jan 2026
- Analyzed 19.4M Medicare provider claims across 2022-2023 and \$182.8B in payments using advanced SQL to identify cost drivers, utilization patterns, and billing anomalies. - Built data-quality checks achieving 100% completeness and 99.9% financial integrity for large-scale healthcare analytics. - Identified \$3.5-4.0B in annual savings opportunities through outlier, specialty-drug growth, and geographic variation analysis.	
From Clicks to Cash - Consulting-Style Ecommerce Analytics (<i>SQL, PostgreSQL, Tableau, Cohort Analysis</i>)	Dec 2025
- Analyzed ~125K orders with SQL cohorts and window functions to diagnose leakage across cancellations, AOV, and repeat-purchase behavior. - Quantified \$23.1M annual revenue upside and delivered Tableau dashboards plus a prioritized growth roadmap for product, growth, and operations teams.	
Efficient LLM Fine-Tuning for Math Reasoning (<i>Python, PyTorch, Hugging Face Transformers, PEFT (LoRA), bitsandbytes</i>)	Nov 2025
- Fine-tuned Microsoft Phi-2 (2.7B) on GSM8K using 4-bit NF4 QLoRA, training only 0.06% of parameters to cut VRAM from ~10GB to ~4GB on a single T4 GPU. - Designed an error-driven training pipeline mining failure cases into a 70/30 weighted dataset, correcting weak reasoning while anchoring against catastrophic forgetting via conservative learning-rate updates. - Ran hyperparameter sweeps and ablation studies comparing baseline vs. error-driven models, achieving 52.9% Exact Match and 0.44 ROUGE-L on multi-step chain-of-thought reasoning tasks.	

Skills

Data & BI: SQL (joins, CTEs, window functions, aggregations), Tableau, Excel (Pivot Tables, XLOOKUP, Power Query, DAX), KPI dashboards, data visualization

Python & ML: Pandas, NumPy, data cleaning, EDA, scikit-learn, XGBoost, regression, classification, Matplotlib, Seaborn

Databases & Cloud: PostgreSQL, MySQL, Redshift, MongoDB, DynamoDB, AWS S3, AWS Textract, Pinecone, Weaviate, FAISS

Analytics & AI Methods: cohort analysis, A/B testing, data quality validation, time-series forecasting, AI response evaluation, RAG workflow testing, requirement analysis

Tools & Practices: Git, Docker, stakeholder communication, storytelling with data, cross-functional collaboration

Licenses & Certifications

- **Fine-tuning Language Models** - Hugging Face, Jun 2025 | Credential ID: Ajay0612
- **Machine Learning, Data Science and Generative AI with Python** - Udemy, Nov 2024
- **The Complete Data Structures and Algorithms Course in Python** - Udemy, Nov 2024